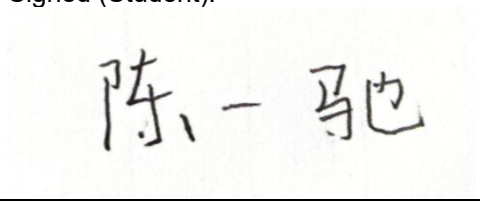
Glasgow College, UESTC

Project Specifications and Preliminary Report on UESTC4006P(BEng) Final Year Project

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Student Name	Chenyichi	Placement Company (if appropriate)	
Student Matriculation Number	[redacted]	Working Title of Project	Research on Deep Learning based Fine Segmentation Technology of Retinal Blood Vessels in the Fundus
UESTC Student Number	[redacted]	Name of First Supervisor	Jiali Yu
Degree programme	Bachelor	**Declaration of Originality and Submission Information**	*I affirm that this submission is all my own work in accordance with the University of Glasgow and UESTC Regulations, and the James Watt School of Engineering requirements.* Signed (Student): 
Academic year	4		

Your report should be NO more than 8 pages in length and include the below subject headings and incorporated within this document:
Project Description (no more than half a page)
Measurable Outcomes (no more than half a page): including Main Tasks and Targets, and Tangible outcomes (Hardware, Software, Hardware & Software, Theoretical research)
Technical Background (at least four pages)**:** including Literature Review, Topic Basis and Significance, and Research Status and Development Trend
References
Work Plan (no more than one page): including Project Outline
Resources: Complete the component request form and email the form to your 1st supervisor separately.
Rist Assessment Form: You must submit a Risk Assessment Form. Please have it signed with your 1st supervisor.
Deadlines for submission of this report: Please upload this report via the Moodle page and GC-UESTC FYP management system by the deadline mentioned in Table 1 of your project handbook.

Project Description

Retinal vessel segmentation is a critical component in the automated screening and diagnosis of ocular pathologies. However, accurately segmenting fine vascular structures remains a significant challenge due to low contrast, noise, and boundary ambiguity, particularly in terminal regions. While many studies have proposed custom-optimized U-Net variants, these solutions often require extensive, dataset-specific manual tuning, which limits their generalizability and reproducibility.

To address these limitations, this project proposes the application and evaluation of nnU-Net, a state-of-the-art, self-configuring deep learning framework. Unlike approaches that rely on manual architectural design, nnU-Net systematically automates the entire experimental pipeline. This process includes preprocessing, network architecture configuration, and training strategy selection to tailor an optimal U-Net-based model specifically for the given dataset.

This research will apply the nnU-Net framework to publicly available retinal datasets (e.g., DRIVE, STARE). The primary objectives are to successfully train and validate the automatically configured models, rigorously evaluate their performance on fine-vessel segmentation, and benchmark the "out-of-the-box" nnU-Net results against

published, state-of-the-art, manually-designed models. This project is expected to yield a robust, reproducible segmentation pipeline and provide a critical analysis of the efficacy of Automated Machine Learning (AutoML) frameworks versus bespoke model design for this specific clinical application.

Measurable Outcomes (no more than half a page)

Main Tasks and Targets:

The primary objective of this project is to leverage the nnU-Net v2 framework to establish a high-performance baseline for retinal vessel segmentation and subsequently develop and evaluate a novel, improved model incorporating a modified loss function.

- Task 1: Baseline Model Establishment. Configure, train, and validate the standard, "out-of-the-box" nnU-Net v2 framework (using its default configuration, which primarily optimizes based on the Dice coefficient) on the target retinal datasets.
- Target 1: Quantify the baseline nnU-Net v2 performance on the DRIVE dataset.
    - *Preliminary Baseline Results (DRIVE Dataset):* The standard nnU-Net v2 framework has been successfully run. Qualitative results (see Figure 1) show effective vessel extraction. The quantitative metrics obtained using the default Dice-based loss configuration are summarized below:

IoU (Vessel)	mIoU	Recall	Precision	Accuracy	mPA	F1-score (Dice)
0.6574	0.8107	0.7688	0.8219	0.9664	0.8768	0.7924

    - These results establish the initial baseline performance.

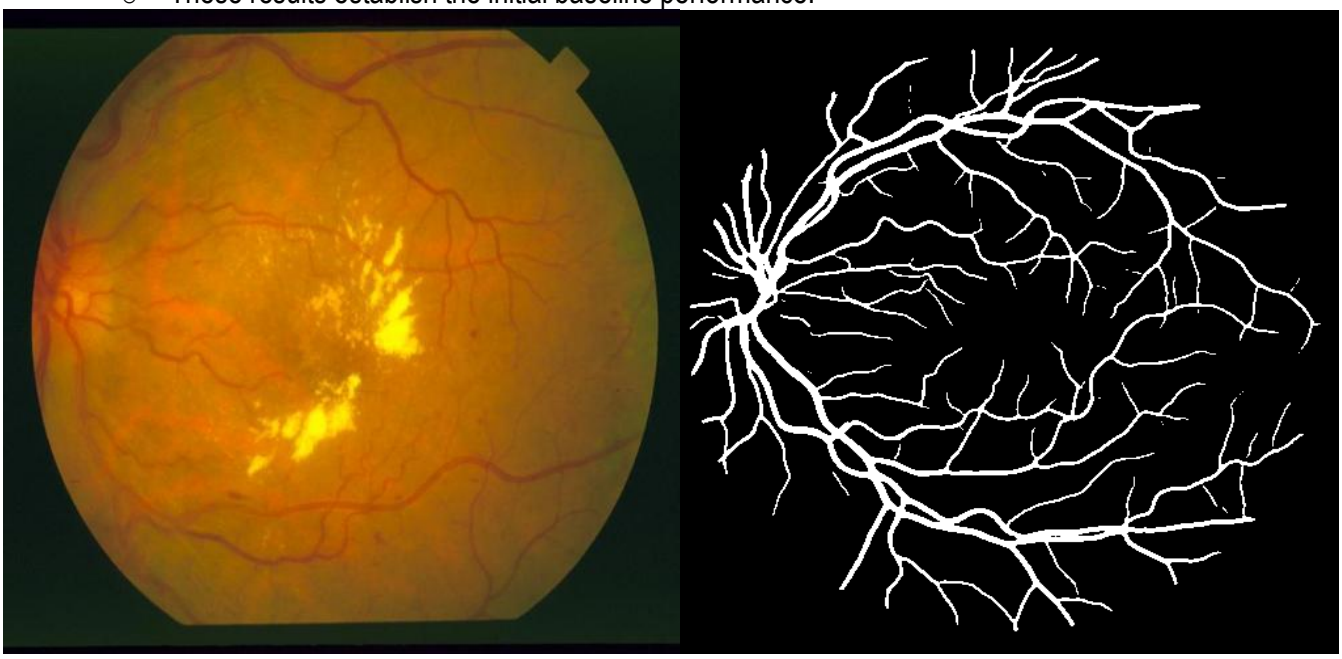


Figure1： Example of baseline nnU-Net v2 segmentation on a sample from the DRIVE dataset.

- Task 2: Development and Evaluation of an Improved nnU-Net Variant. Identify potential limitations of the default Dice loss, particularly for imbalanced classes like thin vessels. Propose, implement, and train a modified nnU-Net variant incorporating a combined Focal Loss and Dice Loss function.
- Target 2: Demonstrate improved performance compared to the baseline, especially for fine vessels. The goal is to evaluate if the modified loss function leads to results closer to or surpassing existing SOTA performance (e.g., F1-score of 0.8505 and IoU of 0.7398 previously cited).
- Analysis. Conduct a rigorous evaluation benchmarking the baseline nnU-Net v2 and the proposed improved model (with modified loss) against other relevant methods reported in the literature.
- Target 3: Produce a final analysis, including quantitative comparison tables and qualitative visualizations, to validate the effectiveness of the modified loss function, particularly in segmenting challenging, fine-vessel regions.

Tangible outcomes:

- This project's primary outcome is a software-based solution. This includes:
  - The fully trained and validated baseline nnU-Net v2 models (default loss) for the target retinal datasets.
  - The complete source code (Python/PyTorch) implementing the modified loss function within the nnU-Net framework and the corresponding final model weights.
  - A set of reproducible data processing, training, and evaluation scripts required to validate the project's findings.

Technical Background

1. Topic Basis and Significance

The intricate network of blood vessels within the human retina serves as a unique and readily accessible window into the body's microcirculation [1, 2]. As direct extensions of the brain's embryonic diencephalon, the retina and optic nerve share significant anatomical, functional, and immunological characteristics with the central nervous system [1, 2]. This deep connection underpins the principle that alterations observed in the retinal vasculature can act as surrogate markers for changes occurring within the cerebral microvasculature, often preceding clinical symptoms of systemic or neurological diseases [1, 2]. Consequently, the analysis of retinal fundus images has become a critical tool in ophthalmology and related medical fields [1-3].

Quantitative assessment of retinal vessel morphology provides invaluable diagnostic and prognostic information. Geometric features such as vessel diameter, tortuosity (curvature), branching angles, density, and the overall complexity of the branching pattern (often characterized by fractal dimension analysis) are sensitive indicators of underlying pathophysiology [1, 2]. For instance, diabetic retinopathy (DR), a leading cause of preventable blindness globally and a frequent complication of diabetes mellitus, manifests through characteristic vascular changes including microaneurysms, hemorrhages, vessel dilation, and neovascularization [2, 4]. Regular screening and accurate assessment of these vascular abnormalities are imperative for detecting sight-threatening stages like proliferative diabetic retinopathy (PDR) and diabetic macular edema (DME) early enough for timely intervention, thus preventing irreversible vision loss [4]. Similarly, hypertensive retinopathy (HR) presents with signs like generalized or focal arteriolar narrowing, arteriovenous nicking, and hemorrhages, reflecting systemic vascular stress due to high blood pressure and correlating with cardiovascular risk [1, 2]. Beyond these well-established links, emerging evidence strongly connects retinal vascular health to neurological conditions [1, 2]. The shared vascular properties between the eye and brain mean that subtle changes in retinal vessel caliber, density, or complexity might mirror early microvascular damage in the brain associated with neurodegenerative diseases like Alzheimer's disease (AD) or cerebrovascular events such as stroke [1, 2]. Studies utilizing fractal dimension analysis, a measure of pattern complexity, have reported significant associations between a sparser, less complex retinal vascular network (lower fractal dimension) and cognitive impairment, dementia (including AD and vascular dementia), and both acute and chronic stroke settings [1]. Accurate segmentation of the retinal vasculature is the indispensable first step for quantifying these morphological biomarkers [1-3]. Whether calculating vessel diameters, assessing tortuosity, or computing fractal dimensions, the fidelity of the underlying vessel map directly determines the reliability and clinical utility of these subsequent analyses [1].

However, obtaining accurate retinal vessel segmentations poses significant challenges [2, 3]. Manual delineation by trained experts, while serving as the gold standard for creating reference datasets, is exceptionally labor-intensive, time-consuming, costly, and suffers from considerable inter- and intra-observer variability, making it unsuitable for large-scale clinical application or screening programs [1-3]. The inherent complexities of retinal images exacerbate these difficulties [2, 3]. Fundus photographs often suffer from non-uniform illumination across the image, low and variable contrast between fine vessels and the retinal background, image noise introduced during acquisition, and the presence of anatomical landmarks (like the optic disc and fovea) or pathological features (e.g., exudates, drusen, hemorrhages) that can obscure or mimic vessels [1, 3]. The vessels themselves exhibit a wide range of calibers, from major arteries and veins down to tiny capillaries, and display complex, tortuous branching patterns that vary significantly between individuals [2, 5]. Fine vessels, in particular, are notoriously difficult to segment accurately due to their low signal-to-noise ratio [2, 5].

The escalating global prevalence of chronic diseases like diabetes and hypertension significantly increases the demand for efficient and reliable retinal screening [4]. Healthcare systems face a growing burden in managing these conditions and preventing their sight-threatening complications [4, 6]. This context underscores the critical need for automated, accurate, and robust methods for retinal vessel segmentation [1, 4, 6]. Such automated tools promise to alleviate the workload of clinicians, improve diagnostic consistency, enable large-scale screening initiatives, facilitate quantitative tracking of disease progression, and potentially unlock new insights into the systemic and neurological implications of microvascular health reflected in the eye [1-4]. Developing effective algorithms to meet this need has been a central focus of medical image analysis research for decades [1, 4].

2. Literature Review (Evolution of Methods)

The pursuit of automated retinal vessel segmentation has traversed several methodological stages, evolving from traditional image processing techniques to sophisticated deep learning models [1, 7].

Early approaches primarily relied on unsupervised methods, which operate without labeled training examples by exploiting intrinsic vessel characteristics [1]. Matched filtering was a foundational technique, using kernels designed to match the expected cross-sectional profile of vessels (often modeled as Gaussian or its derivatives) [1, 5]. Convolution with these filters enhances vessel-like structures [1, 5]. Gao et al., for example, leveraged the Gaussian intensity profile observation to specifically employ Gaussian matched filters (GMF) for enhancing thin vessels prior to subsequent processing [5]. Variations included using different kernel shapes, multi-scale approaches, or combining filter responses [1]. Morphological processing, utilizing structuring elements to probe image shape and topology, was another common unsupervised strategy, employed for vessel enhancement, skeletonization, or noise removal [1]. Vessel tracking or tracing algorithms attempted to iteratively follow vessel centerlines from seed points, often based on local intensity or orientation information [1]. Model-based methods, such as active contours (snakes) or level sets, aimed to evolve a curve or surface to fit vessel boundaries based on image energy terms [1, 3]. While these unsupervised methods provided valuable initial solutions, they often exhibited limitations. They were typically sensitive to image noise and variations in illumination, struggled to reliably detect very fine or low-contrast vessels, could be easily disrupted by vessel crossings or nearby pathologies, and often required careful parameter tuning for different datasets [1].

Supervised machine learning (ML) methods represented a significant step forward by incorporating knowledge from manually annotated ground truth segmentations [1, 6]. The standard pipeline involved two main stages: feature extraction and pixel classification [1, 6]. Feature extraction relied on designing handcrafted features based on domain knowledge to characterize each pixel or its surrounding patch [1, 5, 6]. These features often included intensity values, responses from various filters (e.g., Gabor wavelets, Hessian-based vesselness measures, matched filters), texture descriptors, or geometric properties related to vessel shape and structure [1, 3, 6]. Subsequently, pixel classification was performed using trained classifiers such as k-Nearest Neighbors (KNN), Support Vector Machines (SVM), Artificial Neural Networks (ANN), AdaBoost, or ensemble methods like Random Forests to label each pixel as either 'vessel' or 'background' based on its feature vector [1, 6]. Supervised ML generally yielded better accuracy than unsupervised methods due to the guidance from labeled data [1]. However, their performance was fundamentally limited by the quality and expressiveness of the handcrafted features [1, 5, 6]. Designing optimal features required significant expertise, was often laborious, and resulted in features that might not generalize well across different image types, resolutions, or pathologies [1]. Furthermore, classifying pixels independently often failed to capture the inherent connectivity and smoothness of vascular structures, leading to noisy or fragmented segmentation outputs [5]. Early ML systems applied to clinical screening, such as for diabetic retinopathy, often demonstrated high sensitivity but suffered from relatively low specificity (e.g., 50-69%), generating numerous false positives that hampered their practical utility and cost-effectiveness [4].

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), marked a revolutionary shift in image analysis, including medical image segmentation [1, 4, 6, 7]. CNNs possess the critical advantage of automatically learning hierarchical feature representations directly from raw pixel data, obviating the need for manual feature engineering [1, 7]. Inspired by the architecture of the mammalian visual cortex, CNNs employ layers of convolutions with learnable filters to detect patterns, activation functions (commonly Rectified Linear Units or ReLUs) to introduce non-linearity, and pooling layers (typically max-pooling) to reduce spatial dimensions and achieve a

degree of translational invariance [1, 7, 10]. Seminal architectures like LeNet-5, AlexNet, VGGNet, and GoogleNet demonstrated the immense power of deep feature learning, achieving breakthrough performance on large-scale image classification benchmarks [1, 7].

Adapting CNNs for dense prediction tasks like segmentation required architectural innovations. Fully Convolutional Networks (FCNs) were among the first successful deep learning models for end-to-end semantic segmentation [3, 7]. FCNs replaced the final fully connected layers found in classification CNNs with convolutional layers and utilized upsampling or transposed convolution layers to generate a pixel-wise prediction map matching the input image size [3, 7]. While enabling end-to-end training, FCNs often produced segmentations lacking fine-grained detail, as significant spatial information was lost during the aggressive downsampling in the initial encoding layers [3, 7, 10].

The U-Net architecture, introduced by Ronneberger et al. specifically for biomedical image segmentation, provided a highly effective solution [7]. U-Net is characterized by its symmetric encoder-decoder structure, visually resembling the letter 'U' [3, 7, 10]. The encoder (contracting path) consists of repeated application of two convolutions (typically 3x3) followed by ReLU and a max-pooling operation for downsampling, progressively capturing context while reducing spatial resolution [7]. The decoder (expansive path) symmetrically consists of an up-convolution (transposed convolution) that doubles the feature map size, followed by concatenation with the correspondingly cropped feature map from the encoder path via skip connections, and then two convolutions followed by ReLU [3, 7, 10]. These crucial skip connections allow the network to combine the high-resolution spatial details learned in the early encoder layers with the abstract semantic information captured in the deeper layers, enabling precise localization of boundaries [3, 7, 10]. U-Net proved remarkably successful, achieving state-of-the-art results even with relatively small training datasets, a common scenario in medical imaging [7]. This data efficiency was partly attributed to the network's design and the effective use of data augmentation techniques, such as elastic deformations [7]. The architecture also incorporated strategies like weighted loss maps to force the network to learn separation borders between touching objects [7]. Its impressive performance and elegant design rapidly established U-Net as a foundational model and the go-to benchmark for a vast range of medical segmentation tasks [3, 7]. For handling volumetric data (e.g., 3D MRI or CT scans), the 3D U-Net variant was developed, straightforwardly replacing all 2D operations (convolutions, pooling, up-convolutions) with their 3D counterparts [7, 8].

3. Research Status and Development Trend

While the standard U-Net provided a significant advancement, its limitations, particularly in segmenting intricate, low-contrast structures like fine retinal vessels or dealing with significant image variability and noise, became apparent [3, 7]. Issues include potential loss of fine-grained details in the deeper layers of the encoder, susceptibility to irrelevant background features or noise, and sometimes an insufficient receptive field to incorporate necessary global context, especially when trained on small image patches necessitated by GPU memory constraints [3, 7]. Consequently, a vast body of work has emerged, focusing on developing U-Net variants that address these shortcomings, shaping the current research landscape and future trends.

One prominent trend involves enhancing the core U-Net architecture by integrating more advanced deep learning modules and concepts [3, 7, 9]. Attention mechanisms, designed to allow the network to selectively weight the importance of different features or spatial locations, have been widely explored [3, 7]. Attention U-Net, for instance, incorporates attention gates (AGs) within the skip connections [7]. These gates learn to filter the features passed from the encoder to the decoder, emphasizing relevant structures while suppressing noise or background clutter [7]. Lv et al. proposed AA-UNet, which employs a dedicated attention module to predict a bounding box encompassing the relevant retinal region, generating a spatial attention mask that guides the segmentation process towards the vessel network [10]. Other forms of attention, such as the Convolutional Block Attention Module (CBAM), which combines channel-wise and spatial attention, have also been integrated into U-Net-like architectures, including modifications to the nnU-Net framework itself, to refine feature representations [7, 9].

Atrous (or dilated) convolutions offer another popular enhancement [3, 7]. By inserting gaps between filter weights, atrous convolutions enlarge the filter's receptive field without increasing the number of parameters or reducing the spatial resolution of the feature map [7]. This allows the network to incorporate broader contextual information, which can be beneficial for understanding larger structures or disambiguating local patterns, and is often applied in the bottleneck or deeper layers of U-Net variants [3, 7].

Inspired by the success of very deep networks in classification, residual connections (from ResNet) [7] and dense connections (from DenseNet) [7] have been incorporated into U-Net blocks. Residual connections, forming ResU-Net, add skip connections *within* convolutional blocks, facilitating gradient flow and enabling the training of deeper architectures that might capture more complex features without performance degradation [3, 7]. Dense connections, where each layer within a block receives inputs from all preceding layers, promote feature reuse and gradient propagation even more aggressively [7]. The U-Net++ architecture represents a significant redesign of the skip pathways, introducing nested and densely connected convolutional blocks between the encoder and decoder to progressively aggregate features at multiple scales and bridge the semantic gap [7].

Other architectural innovations include incorporating multi-scale processing explicitly, for example, through parallel pathways with different filter sizes (inspired by Inception networks) [7] or feature pyramid structures. Recurrent connections (forming Recurrent U-Nets) have been explored to potentially model sequential dependencies or iterative refinement [7]. Generative Adversarial Networks (GANs) have been combined with U-Net, using the U-Net as a generator in an adversarial setup, potentially improving segmentation realism or enabling semi-supervised learning [1, 7]. Furthermore, ensemble methods, combining predictions from multiple U-Net models (e.g., trained on different data folds or with different initializations), and cascaded approaches, where multiple U-Nets are chained together (e.g., a first stage for coarse localization and a second for refinement), are frequently employed to boost overall performance and robustness [7, 8].

Despite the impressive performance achieved by many U-Net variants, a major challenge hindering clinical translation remains the lack of methodological standardization and the persistent need for extensive manual effort in designing, tuning, and validating these models for each specific application [1, 6, 7]. The "zoo" of U-Net variants makes it difficult to fairly compare different proposed methods, as performance gains might be attributable not only to the novel architecture but also to differences in implementation details, pre/post-processing, training strategies (optimizer, learning rate, loss function, data augmentation), or dataset splits [1, 7]. This lack of reproducibility and the high cost associated with adapting models to new datasets highlight a critical bottleneck [1].

The rise of Automated Machine Learning (AutoML), and specifically the nnU-Net framework, represents a pivotal development trend aimed squarely at addressing these issues [6]. AutoML seeks to automate parts of the ML pipeline, including neural architecture search (NAS), hyperparameter optimization, and data augmentation strategy selection [6]. Various NAS techniques and predictor-based methods have been explored [6]. Yang et al.'s T-AutoML demonstrated a comprehensive approach using Transformers to search across architecture, hyperparameters, and data augmentation simultaneously [6]. However, many AutoML approaches still focus only on specific parts of the pipeline [6].

nnU-Net, developed by Isensee et al., stands out in the medical imaging domain for its focus on automating the *entire* segmentation pipeline configuration [8, 9, 7]. Adopting a "no-new-Net" philosophy, it leverages robust implementations of standard 2D U-Net, 3D U-Net, and a 3D U-Net Cascade, eschewing complex architectural novelties [7, 8]. Its core strength lies in its "self-adapting" capability: it automatically analyzes the properties of any new dataset (image sizes, spacings, modalities, etc.) and derives a tailored configuration encompassing preprocessing (resampling, normalization), network topology (depth, features, patch size, batch size, cascade use), training parameters (loss, optimizer, learning rate schedule, extensive augmentation), and inference strategy (sliding window, ensembling, postprocessing) based on predefined heuristics and hardware constraints [8]. This fully automated process eliminates the need for manual tuning and ensures reproducibility [8]. nnU-Net's dominant performance across the diverse datasets of the Medical Segmentation Decathlon challenge established it as the de facto state-of-the-art baseline for many medical segmentation tasks, demonstrating that meticulous pipeline configuration is often paramount [7, 8].

The current development trend involves using nnU-Net as a highly optimized and reproducible baseline, upon which targeted enhancements can be built for specific difficult tasks [8, 9]. While nnU-Net excels at generalization, its standardized nature means it might not always achieve the absolute theoretical maximum performance compared to a highly specialized, manually tuned model [8]. Therefore, research is focusing on strategically integrating advanced techniques *on top of* the robust nnU-Net framework. The work by Fang & Huang [9] exemplifies this, showing performance gains on brain tumor segmentation by adding deep supervision (auxiliary losses at intermediate layers

to improve gradient flow and multi-scale learning) [7, 9] and attention mechanisms (CBAM) (to refine features spatially and channel-wise) [7, 9]. Their work also explored alternative activation functions like SiLU [9].
This hybrid approach—starting with the automatically configured nnU-Net pipeline and then integrating well-motivated, task-specific enhancements—seems highly promising for retinal vessel segmentation. Given the specific challenges of this task (e.g., segmenting fine, low-contrast vessels, handling complex branching), potential improvements built upon nnU-Net could include: integrating enhanced pre-processing steps like specialized filters (e.g., GMF [5]) or CLAHE [1-3, 5]; incorporating attention modules (e.g., CBAM [7, 9] or AGs [7]); exploring refined multi-scale feature aggregation techniques [7]; adapting loss functions to better handle vessel thinness or boundary precision [7]; and utilizing deep supervision [7, 9]. The goal is to combine the standardization and robustness of nnU-Net with targeted innovations to achieve the highest possible accuracy and reliability for segmenting the intricate retinal vasculature, especially fine vessels.

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Work Plan

1. Project Outline

Project Title: Research and Implementation of Retinal Image Segmentation Based on Deep Learning

Project Objectives: This project aims to research and apply advanced deep learning segmentation models (such as Unet, Deeplab, Segformer, etc.) to achieve automatic segmentation of key features (especially blood vessels and optic discs) in fundus retinal images. The main objectives include:

1. Build a unified, multi-source retinal image dataset.
2. Learn and build several mainstream image segmentation models.
3. Apply the selected models to complete the segmentation tasks for retinal blood vessels and optic discs.
4. Improve existing models to enhance segmentation performance.

5. Complete the writing and defense of the graduation thesis.

2. Work Plan

